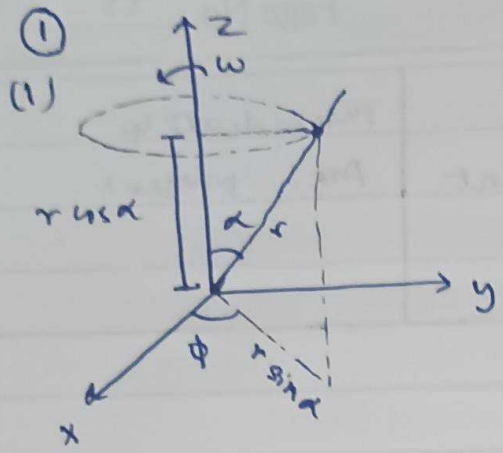
Tutorial - 3 (Classical Mechanics)

① (i)



From the ~~digr~~ diagram;

$$\phi = \omega t$$

$$x = r\sin\alpha \cos\omega t \quad ; \quad z = r\cos\alpha$$

$$y = r\sin\alpha \sin\omega t$$

$$\therefore \dot{x} = \sin\alpha\,[\dot{r}\cos\omega t - \omega r\sin\omega t].$$

$$\dot{y} = \sin\alpha\,(\dot{r}\sin\omega t + \omega r\cos\omega t).$$

$$\dot{z} = \dot{r}\cos\alpha.$$

$$\therefore v^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2$$

$$= \sin^2\alpha\,(\dot{r}^2 + \omega^2 r^2) + \dot{r}^2\cos^2\alpha$$

$$= \dot{r}^2 + \omega^2 r^2 \sin^2\alpha$$

$$\therefore T = \frac{1}{2}mv^2 = \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2\sin^2\alpha)$$

$$\therefore V = mgr\cos\alpha$$

$$\therefore L = T - V = \frac{1}{2}m(\dot{r}^2 + \omega^2 r^2\sin^2\alpha) - mgr\cos\alpha \quad \text{(Ans)}$$

(ii) $$\frac{\partial L}{\partial r} = m\omega^2 r\sin^2\alpha - mg\cos\alpha$$

$$\frac{\partial L}{\partial \dot{r}} = m\dot{r}$$

$$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0 \Rightarrow \boxed{m\ddot{r} = m\omega^2 r\sin^2\alpha - mg\cos\alpha}$$

(Ans)

(iii) $$L = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}m\omega^2 r^2\sin^2\alpha - mgr\cos\alpha$$

$$= \frac{1}{2}m\dot{r}^2 - V_{eff}$$

[We can directly write by comparison as there is no cyclic coordinate; L_z is not conserved]

$$\therefore \boxed{V_{eff} = -\frac{1}{2}m\omega^2 r^2\sin^2\alpha + mgr\cos\alpha} \quad \text{(Ans)}$$

$$\therefore -\frac{dV_{eff}}{dr} = +m\omega^2 r\sin^2\alpha - mg\cos\alpha = F_{eff} = m\ddot{r} \text{ (same EOM).}$$

An analysis of V_{eff} also gives us the same Lagrangian EOM.

For eq^m points ; $\frac{dV_{eff}}{dr} = 0$; $\frac{d^2V_{eff}}{dr^2}$

$\Rightarrow \quad m\omega^2 r \sin^2\alpha - mg\cos\alpha = 0$.

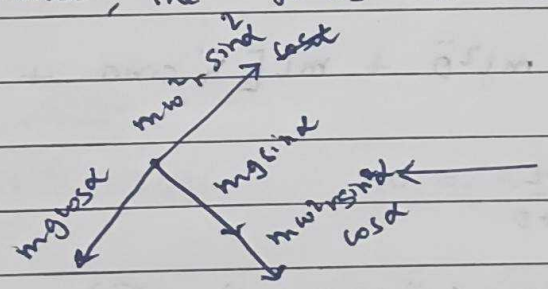
$\Rightarrow \quad \boxed{r_{eq} = \frac{g\cos\alpha}{\omega^2 \sin^2\alpha}}$

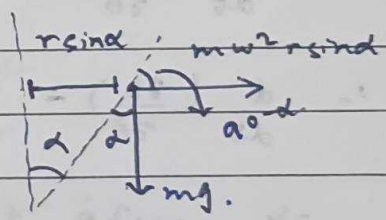
$\therefore \frac{d^2V_{eff}}{dr^2} = -m\omega^2 \sin^2\alpha < 0$.

Thus, it is an unstable eq^m.

Interpretation:

From the rod's frame, the forces on the bead are:





The forces along the wire are in opposite directions.

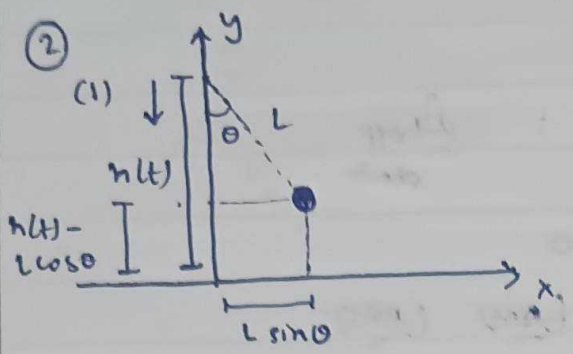
@ eq^m:

$$mg\cos\alpha = m\omega^2 r\sin^2\alpha.$$

$$\Rightarrow \quad \boxed{r_{eq} = \frac{g\cos\alpha}{\omega^2\sin^2\alpha}}$$

But it is an unstable eqm. Any perturbation, would remove the bead from the eqm ~~position~~ position.

(2)



$\therefore\ x = L\sin\theta \quad ; \quad y = h(t) - L\cos\theta$

$\dot{x} = L\cos\theta\,\dot{\theta} \quad ; \quad \dot{y} = \dot{h}(t) + L\sin\theta\,\dot{\theta}$

$v^2 = \dot{x}^2 + \dot{y}^2$

$= L^2\dot{\theta}^2 + \dot{h}^2 + 2\dot{h}L\sin\theta\,\dot{\theta}$

$T = \frac{1}{2}mv^2 = \frac{1}{2}m(L^2\dot{\theta}^2 + \dot{h}^2 + 2\dot{h}L\sin\theta\,\dot{\theta})$

$V = mgy = mg[h - L\cos\theta]$

$\therefore\ L = T - V = \frac{1}{2}m(L^2\dot{\theta}^2 + \dot{h}^2 + 2\dot{h}L\sin\theta\,\dot{\theta}) - mg(h - L\cos\theta)$

(Ans)

(ii) $\frac{\partial L}{\partial \theta} = m\dot{h}L\dot{\theta}\cos\theta - mgL\sin\theta \quad ; \quad \frac{\partial L}{\partial \dot{\theta}} = mL^2\dot{\theta} + m\dot{h}L\sin\theta$

$\therefore\ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = mL^2\ddot{\theta} + mL[\ddot{h}\sin\theta + \dot{h}\cos\theta\,\dot{\theta}]$

$\therefore\ \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$

$\Rightarrow\ mL^2\ddot{\theta} + mL(\ddot{h}\sin\theta + \cancel{\dot{h}\cos\theta\,\dot{\theta}}) - \cancel{m\dot{h}L\dot{\theta}\cos\theta} + mgL\sin\theta = 0$

$\Rightarrow\ \boxed{mL^2\ddot{\theta} + mL\ddot{h}\sin\theta + mgL\sin\theta = 0}$

(iii) The EOM is:

$mL^2\ddot{\theta} + mL\sin\theta(\ddot{h} + g) = 0.$

If $\ddot{h} = -g \implies mL(\ddot{h} + g)\sin\theta = 0.$

Reduced EOM : $m l^2 \ddot{\theta} = 0$

$\Rightarrow \ddot{\theta} = 0 \Rightarrow \theta(t) = At + B$

$B = \theta(0) \longrightarrow$ Initial pendulum angle.

$A = \dot{\theta} \longrightarrow$ Angular velocity.

In this case of free-fall, the pendulum rotates around the pivot with constant angular ~~velo~~ velocity.

$\ddot{h} = -g \Rightarrow \dot{h} = -gt + a$

$\Rightarrow \boxed{h(t) = -\frac{gt^2}{2} + at + b}$

If $h(t)$ is of the form given, then g can be compensated by $h(t)$.

(3) - Using polar coordinates in 2D :

$v = \dot{r}^2 + r^2\dot{\theta}^2$

$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) \quad ; \quad V = -\frac{e^{-\alpha r}}{r}$

$\therefore L = T - V = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{e^{-\alpha r}}{r}$

$= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + \cancel{\frac{e^{-\alpha r}}{r}} \; V \; \frac{e^{-\alpha r}}{r}$

$\cancel{= \frac{1}{2}m\dot{r}^2 \quad V_{eff}(r)}$

$\therefore V_{eff} = \cancel{\frac{e^{-\alpha r}}{r} + \frac{1}{2}mr^2\dot{\theta}^2}$ $\qquad \dot{\theta} = \frac{L_z}{mr^2} = \frac{h}{r^2}$

$= -\frac{e^{-\alpha r}}{r} + \frac{mh^2}{2r^2}$

Teacher's Signature : _____

(See the Last section to know how we get it)

$\therefore \frac{\partial L}{\partial r} = mr\dot{\theta}^2 \mp \frac{e^{-\alpha r}}{r}\left(\frac{1}{r}+\alpha\right) = \frac{mh^2}{r^3} \mp \frac{e^{-\alpha r}}{r}\left(\alpha+\frac{1}{r}\right).$

$\therefore \frac{\partial L}{\partial \dot{r}} = m\dot{r}$

$\therefore \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0$

$\Rightarrow \boxed{m\ddot{r} = \frac{mh^2}{r^3} \mp \frac{e^{-\alpha r}}{r}\left(\frac{1}{r}+\alpha\right)}$ — The same EOM was obtained by an analysis of V_{eff}.

Eq^m pts. @ $\frac{\partial V_{eff}}{\partial r} = 0$: $\left[\frac{\partial V}{\partial r} = -\frac{e^{-\alpha r}}{r}\left(\alpha+\frac{1}{r}\right); \quad \frac{\partial}{\partial r}\left(\frac{mh^2}{2r^2}\right) = -\frac{mh^2}{r^3}\right]$

$\frac{e^{-\alpha r}}{r}\left(\frac{1}{r}+\alpha\right) = +\cancel{mr\dot{\theta}^2}\ \frac{mh^2}{r^3}.$

$\Rightarrow e^{-\alpha r}\left(\alpha+\frac{1}{r}\right) = \cancel{mr^2\dot{\theta}^2} + \frac{mh^2}{r^2}$

$\left[\frac{\partial V_{eff}}{\partial r} = -\frac{e^{-\alpha r}}{r}\left(\alpha+\frac{1}{r}\right) - \frac{mh^2}{r^3}\right]$

RHS: $+\frac{mh^2}{r^2} > 0$.

LHS: $e^{-\alpha r} > 0$; $\alpha > 0$

$\Rightarrow \alpha + \frac{1}{r} > 0$ [as $r > 0$].

$\Rightarrow e^{-\alpha r}\left(\alpha+\frac{1}{r}\right) > 0$

$\therefore$ ~~NOT possible~~ RHS > 0 ; LHS > 0 $\longrightarrow$ ~~Not possible~~ Possible!

~~Hence, no~~ Eq^m points exist!

Date.....

Expt. No.....

Page No.....

$\cancel{m\ddot{r}} = \dot{\theta} = \frac{L_z}{mr^2}$ [θ is a cyclic coordinate].

$$\therefore m\ddot{r} = mr \cdot \frac{L_z^2}{m^2r^4} \mp \frac{e^{-\alpha r}}{r}\left(\alpha + \frac{1}{r}\right)$$

$$= \frac{L_z^2}{mr^3} \mp \frac{e^{-\alpha r}}{r}\left(\alpha + \frac{1}{r}\right)$$

$$= -\frac{\partial}{\partial r}\left(\frac{L_z^2}{2mr^2} \mp \frac{e^{-\alpha r}}{r}\right) = -\frac{\partial V_{eff}}{\partial r}$$

$$\therefore -\frac{\partial V_{eff}}{\partial r} = m\ddot{r} \quad \text{—— EOM.}$$

$$V_{eff} = \frac{L_z^2}{2mr^2} \mp \frac{e^{-\alpha r}}{r} = -\frac{e^{-\alpha r}}{r} + \frac{mh^2}{2r^2} \quad \text{(Ans)}$$